BMS INSTITUTE OF TECHNOLOGY AND MANAGEMENT

Department of MCA · Project Work Phase I · MMCA34 · 2025-27

Incident Analysis and Hotspot Identification for Security Provision Against Women Harassment in Public Spaces

SafeHer - crowdsourced reporting, geospatial analysis, self-serve authority dashboard

Anusha K 1BY25MC017 Module 1 Analysis and hotspot identification

Aman Yadav 1BY25MC012 Module 2 Mobile application and dashboard

Guide: Dr. M Sridevi SDG 5 Gender Equality, Target 5.2

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Sequence of this review

01 Organisation - one project, two modules, who builds what.

02 Problem and objectives - the gap, and what we set out to do.

03 End-to-end flow - how a report becomes a map for an officer.

04 Module 1 method - clean data, group nearby reports, score risk.

05 Authority dashboard - maps, filters and a patrol list they open themselves.

06 Security and results - who can see what, and the Karnataka numbers.

PROJECT ORGANISATION

One system. Two complementary modules.

Module 1 (Anusha K, 1BY25MC017): incident analysis, geospatial clustering, hotspot identification. Folder Module1_Analysis/. Command python3 Module1_Analysis/pipeline.py.

Inputs: app reports, OSM area scores, KSP district context. Outputs: incident view, hotspots, area risk, patrol alerts. Rule-based. Not crime prediction.

Module 2 (Aman Yadav, 1BY25MC012): Expo mobile application and Next.js authority dashboard at /admin.

Authorised officers retrieve maps, filters and alerts themselves. They do not wait for another desk to send a file.

PROBLEM STATEMENT

Concentrated locations are not available in time

Incidents reach stations through registers, helplines or informal messages. A duty officer often depends on another person for the latest list.

Need: GPS reporting from a telephone; analysis into hotspots with published rules; one dashboard that authorities operate themselves.

Scope: historical concentration, not prediction. No live CCTV. No facial recognition. KSP tables are district context only.

OBJECTIVES

Aligned to SDG 5, Target 5.2

Collect location, time, incident class and street context from the mobile application over HTTPS.

Store approved reports with Row Level Security so the public cannot read rejected rows.

Analyse: clean, classify, cluster on a ~1 km grid, assign Low / Medium / High.

Alert: patrol list for High-risk or recently active cells, with area, coordinates and last seen.

Present a login-protected dashboard with filters, map layers and KPIs.

Protect: anonymous-first reporting; closed command view.



DATA GATHERING

The telephone is the sensor

Each successful submit starts the pipeline

Location - GPS latitude and longitude, matched to one of 30 Karnataka districts.

Classification - Safe / Unsafe. Incident-like = unsafe, harassment flag, or incident type.

Time - server timestamp plus morning / afternoon / evening / night.

Street context - lighting, crowd, CCTV seen, police nearby, transport, optional comment.

Identity - name is not required. Device identifier only, to cap five reports per hour.

Transport - HTTPS insert into Supabase reports. Default status approved; may be held pending.

In plain words

Only approved rows can be read.

03 **Live aggregation**

In plain words

Counts and the map refresh by themselves.

In plain words

Nearby reports are grouped on a 1 km grid.

03 **Publish artefacts**

In plain words

JSON / CSV / map files update the dashboard.

05

Hotspot detection

In plain words

A square with 3 or more reports is a hotspot.

06

Risk stratification

In plain words

Colour it High, Medium or Low using published rules.

07

Alert generation

In plain words

High or recently active squares become the patrol list.

08

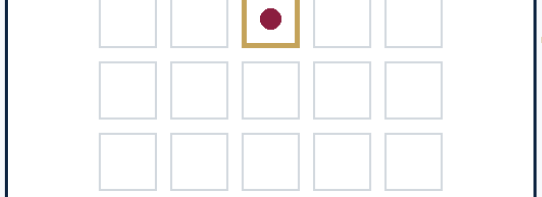
Serving layer

In plain words

Save JSON, CSV and a map. The dashboard reads them.



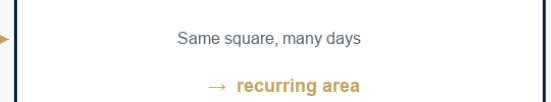
Each pin is one report with GPS.



Pins snap to a 0.01° cell (~1 km).



Keep cells with at least 3 reports.



Reports on 2+ different days = repeating.

ANALYSIS ARTEFACTS

All required files live in Module1_Analysis/

OSM dumps and camera inventories are excluded

Input: Karnataka OSM District Summary (604,573 streets, 30 districts), reports.json (1,079 demo), live_reports.json, crime_context.json (KSP, all districts).

Output: incident_view.json, hotspots.json, alerts.json, areas.json, summary.json, pipeline_log.json.

Dashboard copies: summary, hotspots, alerts, areas under web/public/data/analysis/.

Medium

Else if count ≥ 4 (and not High).

In plain words

Watch this square.

Low

Else the cell is still a hotspot, but quieter.

In plain words

Note it; lower priority.

CURRENT ANALYSIS RESULT

Descriptive statistics · Karnataka · not a forecast

1079	534	72	55
Reports	Incident-like	Hotspots	Alerts

Risk split: High 30 Medium 19 Low 23

Highest incident-like districts: Kolar 31, Bengaluru Urban 27, Mandya 26, Tumakuru 26, Belagavi 25.

Time of day is even - morning 143, afternoon 140, evening 119, night 132. Type mix: unsafe feeling 287, harassment 247.

OSM: 604,573 Karnataka streets, avg score 4.16. Preprocess dropped 0 duplicates.

Live rows in Module1_Analysis/input/live_reports.json use the same pipeline.

SELF-SERVE FOR AUTHORITIES

Officers do not depend on anyone to send them data

Independent access - login at /admin; maps, KPIs, hotspots and alerts are already loaded.

No request to another desk - filters replace a request for last week's spreadsheet.

One operational picture - OSM streets, community reports and analysis circles.

Actionable - High-risk cells carry area, coordinates and last seen.

Stable rules - risk comes from the published pipeline. Each hotspot stores count, type mix, peak time, first and last seen.

AUTHORITY WORKFLOW

Intended morning use

01 Authenticate - sign in at /admin. HttpOnly cookie. Sign-out clears the session.

02 Read the KPIs - hotspot count, High-risk cells, alert count.

03 Filter - area, incident type, from/to date, time of day, risk. Table and map follow.

04 Brief and act - inspect High circles; copy patrol hints into the day's briefing.

In plain words

We store where it happened, not a tracking trail.

04 Row Level Security

In plain words

The public may read only approved rows.

In plain words

Every /admin page except login needs a session.

04 No reporter identity

In plain words

Officers see places and counts, not people.

SECURITY CONTROLS IMPLEMENTED

Present in the current codebase

No name, telephone or photograph required. Device identifier only, to cap five reports per hour. HTTPS to Supabase.

Row Level Security: insert allowed; public SELECT only where status = approved. Rejected rows never map.

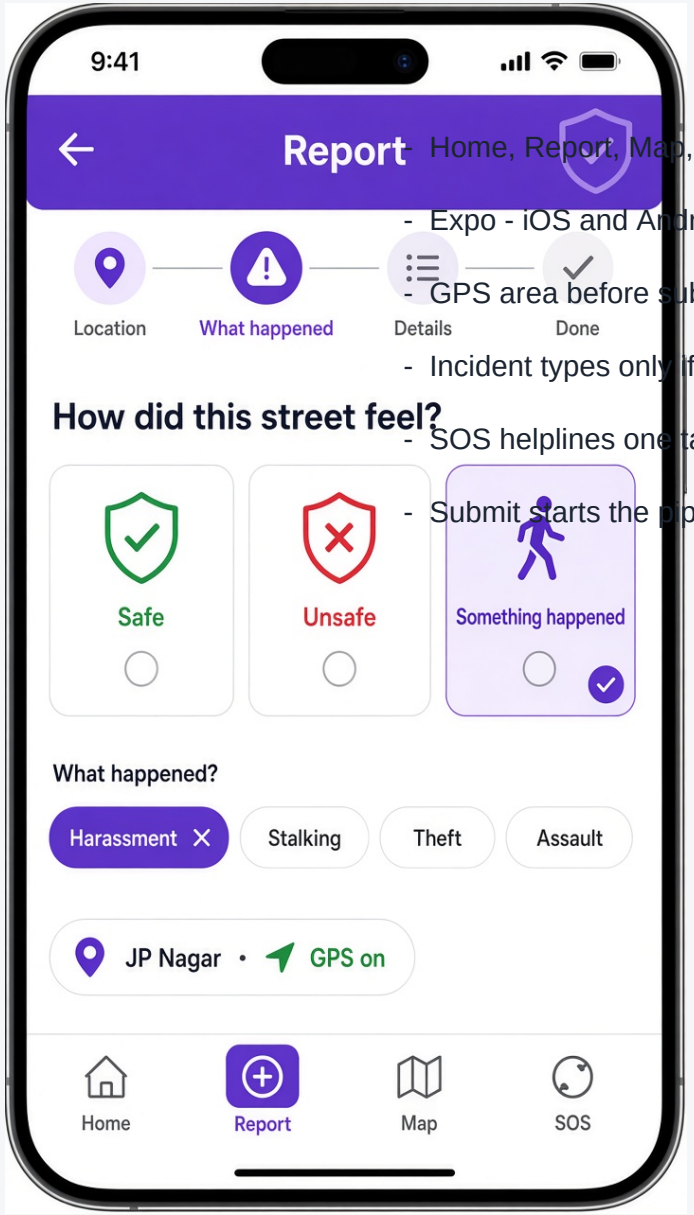
GPS places the report. No tracking trail of the reporter is retained.

Dashboard: middleware blocks /admin without a valid session. Password checked on the server. ADMIN_PASSWORD is an environment variable.

Cookie: HttpOnly, SameSite=Lax, Secure in production, seven-day expiry. Sign-out deletes the session.

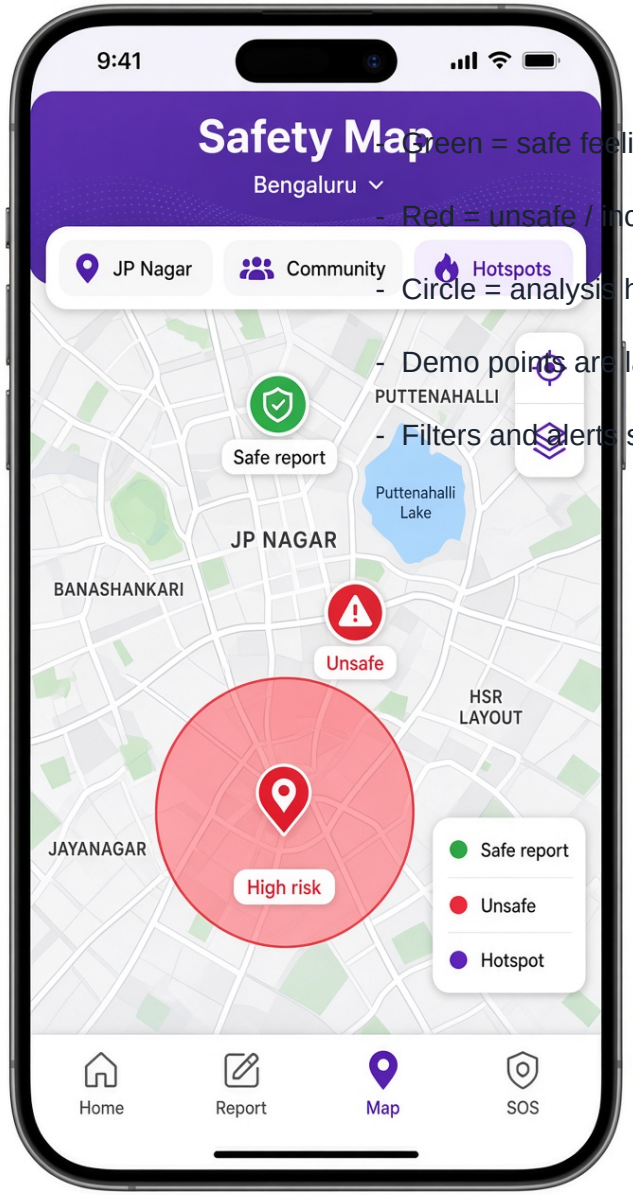
Officers see locations and aggregates. Identities are not stored and therefore cannot be disclosed.

Location, classification, details, confirmation



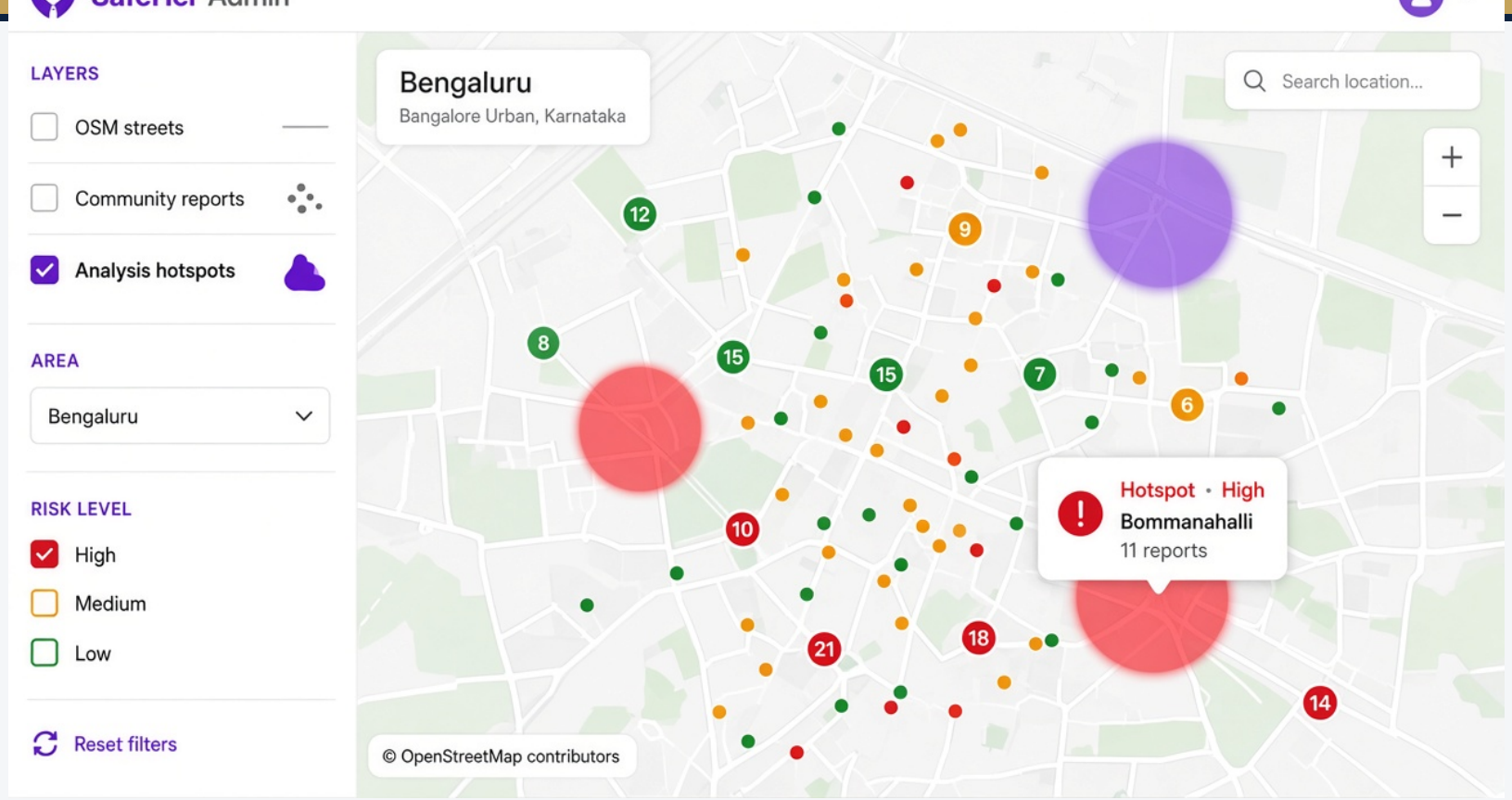
- Home, Report, Map, SOS
- Expo - iOS and Android
- GPS area before submit
- Incident types only if needed
- SOS helplines one tap away
- Submit starts the pipeline

Residents see community signal; command view is on /admin



- Green = safe feeling
- Red = unsafe / incident
- Circle = analysis hotspot
- Demo points are labelled
- Filters and alerts stay on /admin

Grey rows fall outside the current filter and are not shown to the officer.



- Toggle each layer
- Same filters as the table
- Click: count, peak time, risk
- Patrol hint for the briefing
- No GIS export required

What each layer is, and what it is not

OSM streets and camera locations - observed geography (604,573 Karnataka streets, 30 districts).

Street safety score - modelled nine-factor proxy, not a crime rate.

This run's reports - 1,079 synthetic rows across 30 districts; live uses the same table and pipeline.

Hotspots - derived 1 km clusters. Historical concentration, not a forecast.

KSP tables - official, district grain, never street dots.

CCTV video - out of scope.

STATUS AT INTERIM

Process defined and implemented for local demonstration

Completed: synopsis, Karnataka OSM dataset, analysis self-check (72 hotspots, 55 alerts across 30 districts).

Current numbers: 1,079 reports -> 534 incident-like -> 72 hotspots (30 High) -> 55 alerts.

Remaining: live Expo/Next in the laboratory; optional timer to refresh analysis; optional pending-moderation.

Public store deployment is optional and not required to demonstrate the process.

Thank you

Authorities retrieve maps, filters and alerts themselves - without depending on anyone to send the data.

Anusha K · 1BY25MC017 Aman Yadav · 1BY25MC012

Guide: Dr. M Sridevi Department of MCA, BMS Institute of Technology and Management